

UBT610: INDUSTRIAL BIOTECHNOLOGY

L	T	P	Cr
3	0	0	3.0

Course Objective: To understand the use of living cells such as bacteria, yeast, algae or component of cells like enzymes, plants and animals to generate industrial products and processes. To study techniques for genetic improvement of micro-organisms to improve yield of bioproducts.

Syllabus

Introduction to Industrial Biotechnology: Overview of fermentation and other bioproducts Industries. Industrial Bioproducts and their producer organisms. Characteristic properties of Industrially important strains.

Strain improvement in industrial microorganisms: Improvement through mutation and recombination, Integrated Strain improvement program, targets for industrial strain improvement, Increase product concentration, Process improvement, biosynthesis of new metabolites. Precision Engineering Technology, biosynthetic technology.

Process technology for the industrial production of cell biomass, primary and secondary metabolites: SCP, Ethanol, Citric acid, Dextran and Amino acids. (Glutamic acid, L-Lysine.) SCP, beer and vinegar; bio preservatives (Nisin), cheese, biopolymers (xanthan gum, PHB, etc.).

Industrial Production: Vitamin (E, K, B₂ and B₁₂), antibiotics, microbial production of Bio pigments and flavours.

Production of enzymes and Biotransformation: Production of industrial enzymes (proteases, amylases, lipases, cellulases), whole cell biocatalysts, Applications of bioconversion, transformation of steroids and sterols; Microbially enhanced oil recovery (MEOR) and Bioleaching.

Laboratory Work (if applicable):

Isolation of amylolytic microorganisms; Production and partial purification of Amylase in shake flask culture, Production of Citric acid using *Aspergillus* species, Strain improvement of *Aspergillus* species using physical mutagenesis process; Strain improvement of *Aspergillus* species using chemical mutagenesis; Screening and isolation of cellulase producing enzymes; Determination of cellulolytic activity by DNS method; Screening microorganism for antibiotic production, Production and partial purification of Penicillin.

Approved in 1st meeting of the Academic Council held on September 11, 2025. Further revision approved in 3rd meeting of the Academic Council held on March 18, 2026, incorporating modification in credits due to the addition of AI courses.

Course Learning Objectives (CLO)

The students will be able to:

1. Comprehend role of industrial biotechnology in improving microbial cells as factories.
2. Know the production aspects of commodity chemicals, pharmaceuticals and fine chemicals.
3. Apply knowledge of microorganisms in commercial production of flavours, fragrance, and microbial pigment in textile and industry.
4. Apply the process for commercial production of enzyme.
5. Know the process of Microbial Enhanced Oil Recovery and Microbial Leaching.

Text Books

1. Lee, S.Y., Nielsen, J. and Stephanopoulos, G., "Industrial Biotechnology: Products and Processes", John Wiley & Sons, 2016.
2. Waites, M.J., Morgan, N.L., Rockey, J.S., Highton, G., "Industrial Microbiology: An Introduction" Blackwell, 2001.
3. Cruger, W., Cruger, A., "A Textbook of Industrial Microbiology", Panima Publishing Corporation, 2nd Edition, 2005.

Reference Books

1. Pandey, A., Negi, S., Soccol, C.R., "Current Developments in Biotechnology and Bioengineering: Production, isolation and purification of industrial products", Elsevier, 2016.
2. Okafor, N., "Modern Industrial Microbiology and Biotechnology", CRC Press, 2007
3. Prescott and Dunn's "Industrial Microbiology", CBS Publisher, 1987.
4. Casida Jr, L. E., "Industrial Microbiology", Wiley, 1968.

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	MST	25-35
2.	EST	35-45
3.	Sessional (May include Assignments/ Quizzes)	30

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